Stochastic optimal control

Narek Sargsian, Ivan Khromyshchenko

HSE FES Probability Theory Club

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Introduction

- Motivation for the use of stochastic optimal control
- Tasks that can be solved using stochastic optimal control
- Necessary tools
- An optimal periodic dividend and risk control problem for an insurance company
- Merton's Optimal investment problem
- Next steps

Motivation

Deterministic optimal control assumes the system evolves predictably. In practice, however, many processes are stochastic. Incorporating randomness into the control framework therefore makes models more realistic.

Problem scope overview

- Insurance company problem
- Merton's Optimal investment problem
- Passport options
- Avellaneda-Stoikov model

HJB

Define the value function (cost minimization):

$$V(t,x) = \min_{u(\cdot)} \left[\int_t^\tau C(s, x(s), u(s))\, ds + D(x(\tau)) \right] \tag{1}$$

subject to the **state dynamics**:

$$\dot{x}(s) = f(s, x(s), u(s)), \qquad x(t) = x. \tag{2}$$

where:

- $C(t,x,u)$ is the **running cost** (cost rate over time),
- $D(x)$ is the **terminal cost** at the horizon τ.

HJB

The HJB equation is

$$\partial_t V(t,x) + \min_u \{\partial_x V(t,x) f(t,x,u) + C(t,x,u)\} = 0, \tag{3}$$

with terminal condition

$$V(\tau, x) = D(x). \tag{4}$$

Stochastic HJB

In the stochastic case, define the value function (cost minimization):

$$V(t,x) = \min_{u(\cdot)} \mathbb{E}_{t,x}\left[\int_t^\tau C(s, X_s, u_s)\, ds + D(X_\tau)\right], \tag{5}$$

where $\mathbb{E}_{t,x}$ denotes conditional expectation given $X_t = x$.
The controlled state process satisfies

$$dX_s = f(s, X_s, u_s)\, ds + \sigma(s, X_s, u_s)\, dW_s, \qquad X_t = x. \tag{6}$$

Stochastic HJB

The HJB equation is

$$\partial_t V(t,x) + \min_u \{\mathcal{L}^u V(t,x) + C(t,x,u)\} = 0, \tag{7}$$

with terminal condition $V(\tau, x) = D(x)$.
where

$$\mathcal{L}^u V(t,x) = \sum_{i=1}^{d} b_i(t,x,u)\,\partial_{x_i} V(t,x) + \frac{1}{2}\sum_{i,j=1}^{d} a_{ij}(t,x,u)\,\partial^2_{x_i x_j} V(t,x),$$

where $b(t,x,u) = f(t,x,u)$ and $a(t,x,u) = \sigma(t,x,u)\sigma(t,x,u)^\top$.

Transition to stationarity

If the problem is time-invariant (time-homogeneous dynamics and costs), the value function does not depend explicitly on time:

$$V(t, x) = V(x).$$

Therefore,

$$\partial_t V(t, x) = 0,$$

and the HJB equation becomes stationary.

Algorithm

- Defining the dynamics of the process (SDE)
- Defining the objective functional and value function
- Admissible control set
- Hamilton–Jacobi–Bellman (HJB) equation
- First-order conditions and optimal control

Motivation

Before Merton, portfolio theory relied on the Markowitz model. The latter calculates weights only once. Merton's model chooses a dynamic policy.

Markowitz's Model in the utility form:

$$\max_{w \in \mathbb{R}^n} \; w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1$$

- n is the number of risky assets.
- $R \in \mathbb{R}^n$ is the random vector of single-period risky-asset returns.
- $w \in \mathbb{R}^n$ is the vector of portfolio weights in the risky assets.
- $\mu \in \mathbb{R}^n$ is the vector of expected returns, $\mu = \mathbb{E}[R]$.
- $\Sigma \in \mathbb{R}^{n \times n}$ is the covariance matrix of returns, $\Sigma = \mathrm{Cov}(R)$.
- $w^\top \mu$ is the portfolio expected return.
- $w^\top \Sigma w$ is the portfolio return variance.
- $\gamma > 0$ is the risk-aversion parameter (larger γ puts more penalty on variance).

Informal Problem Statement

The investor chooses how much to consume over time c_t and how to allocate wealth between risky assets and the risk-free asset.

Problem Notation

The utility function is of the constant relative risk aversion (CRRA) form (Merton's parameterization):

$$u(x) = \frac{x^\gamma}{\gamma}, \qquad \gamma < 1,\ \gamma \neq 0. \tag{8}$$

(Then relative risk aversion is $1 - \gamma$.)
A utility function other than CRRA can be used.

- $\mathcal{A}$ is the set of strategies $a = (\pi, c)$ and τ_a is the stopping time.
- X_t is the wealth at time t
- The initial wealth is x

Problem Statement

The objective is

$$J^T(x;\pi,c) = \mathbb{E}\left[\int_0^T e^{-\rho s}\, u(c_s)\, ds \;+\; e^{-\rho T}\, u(X_T)\,\middle|\, X_0 = x\right] \tag{9}$$

where the wealth dynamics follow the SDE

$$dX_t = [(r + \pi_t(\mu - r))X_t - c_t]\, dt + X_t \pi_t \sigma dW_t \tag{10}$$

- $\rho > 0$ is the subjective discount rate
- μ is the expected return of the risky asset
- σ is the volatility of the risky asset
- r is the risk-free rate

Defining the value function

Assuming $T \to \infty$ (and a transversality condition so the terminal term vanishes), define the infinite-horizon objective

$$J(x;\pi,c) = \mathbb{E}\left[\int_0^\infty e^{-\rho s}\, u(c_s)\, ds \,\middle|\, X_0 = x\right]. \tag{11}$$

The value function is

$$V(x) = \sup_{(\pi,c)\in\mathcal{A}} J(x;\pi,c). \tag{12}$$

Stationarity (infinite horizon)

With time-homogeneous coefficients and an infinite horizon, the value function is stationary: $V(t, x) = V(x)$. The HJB equation therefore reduces to an ODE in wealth x.

HJB equation

The optimality equation becomes an ODE in wealth:

$$0 = \max_{c,\pi} \Big[u(c) - \rho V(x) + V'(x)\big((\pi(\mu - r) + r)x - c\big) + \frac{1}{2} V''(x)\sigma^2\pi^2 x^2 \Big]. \tag{13}$$

Solution

The first-order conditions are

$$0 = u'(c) - V'(x), \tag{14}$$

$$0 = (\mu - r)V'(x) + V''(x)\sigma^2 \pi x. \tag{15}$$

Solution

The optimal decision rules are:

$$\boxed{\pi^*_\infty = \frac{\mu - r}{\sigma^2(1-\gamma)}} \tag{16}$$

$$\boxed{c^*_t = \left\{ \frac{\rho}{1-\gamma} - \gamma \left[\frac{(\mu - r)^2}{2\sigma^2(1-\gamma)^2} + \frac{r}{1-\gamma} \right] \right\} X_t} \tag{17}$$

Motivation

The surplus of an insurance company can be modeled as a stochastic process. This allows us to apply stochastic optimal control methods to the problem of managing the reinsurance part and dividend payments.

Problem Statement

The surplus process can be described as follows

$$X_t = x_0 + \lambda a[1+\eta]t - \lambda a[1-u_t][1+\mu]t - u_t \sum_{j=1}^{N_t^\lambda} Y_j - L_t, \quad \text{for } t \geq 0, \quad (18)$$

- x_0 denotes the initial surplus of the insurance company.
- u_t represents the proportion of risk transferred to the reinsurer at time t.
- Y_j is the size of the j-th insurance claim, where $\mathbb{E}[Y] = a$ and $\mathrm{Var}(Y) = \widehat{\sigma}^2$.
- λ is the intensity (arrival rate) of insurance claims.
- η denotes the premium inflow rate from policyholders.
- μ is the payment rate from the insurer to the reinsurance company.
- L_t is a non-decreasing process representing the cumulative dividends paid up to time t.

Problem Statement

We can rewrite the process as

$$X_t = x_0 + \lambda a\big[\eta - [1-u_t]\mu\big]t + u_t \underbrace{\left[\lambda a t - \sum_{j=1}^{N_t^\lambda} Y_j\right]}_{Z_t} - L_t, \quad \text{for } t \geq 0.$$

We have $\mathbb{E}[Z_t] = 0$ and $\mathrm{Var}(Z_t) = \lambda t(\widehat{\sigma}^2 + a^2) \approx \widehat{\sigma}^2 \lambda t$. For $\lambda \gg 1$, a diffusion approximation yields $Z_t \approx \sigma W_t$, where W_t is a standard Brownian motion and $\sigma = \widehat{\sigma}\sqrt{\lambda}$.

Dynamics of the process and defining the dividend payment

The diffusion approximation is

$$dX_t = [\eta - [1 - u_t]\mu]dt + \sigma u_t dW_t - dL_t, \quad \text{for } t \geq 0 \text{ and } X_0 = x_0. \quad (19)$$

Dividend payments may be described as follows

$$L_t = \int_0^t \nu_s \, \mathrm{d}N_s^\gamma, \quad \text{for } t \geq 0, \quad (20)$$

where $\gamma > 0$ is the intensity.

Defining the set of controls

Let $\mathcal{A}$ denote the set of strategies $\pi = (u^\pi, L^\pi)$ such that $u^\pi \in [0,1]$, L^π satisfies (20), and $0 \leq \nu_t^\pi \leq X_{t-}^\pi$. A strategy π is said to be admissible if $\pi \in \mathcal{A}$.

Defining NPV

For each $\pi \in \mathcal{A}$, the expected NPV is given by

$$J^{\pi}(x) = \mathbb{E}_x \left[\int_0^{\tau^{\pi}} e^{-\delta t} dL_t^{\pi} \middle| X_0 = x \right] = \mathbb{E}_x \left[\int_0^{\tau^{\pi}} e^{-\delta t} \nu_t^{\pi} dN_t^{\gamma} \middle| X_0 = x \right] \tag{21}$$

Here, $\mathbb{E}_x$ denotes the conditional expectation given the initial surplus level x, $\delta > 0$ is the discount rate, and $\tau^{\pi} := \inf\{t > 0 : X_t^{\pi} < 0\}$ represents the time of ruin. Our objective is to determine the value function V defined by

$$V(x) := \sup_{\pi \in \mathcal{A}} J^{\pi}(x), \qquad x \geq 0. \tag{22}$$

There is an assumption that $V \in C^2(0, \infty)$ and concave.

HJB equation

To find the optimal V, we construct the HJB equation as follows:

$$V(x) = \underbrace{(\nu + \mathbb{E}_x[V(X_t - \nu)])\gamma t e^{-\delta t}}_{\text{in case of dividend payment}} + \underbrace{(1-\gamma t)e^{-\delta t}\mathbb{E}_x[V(X_t)]}_{\text{in case no dividend is paid}} + o(t). \tag{23}$$

$$(\nu + \mathbb{E}_x[V(X_t - \nu) - V(x)])\gamma t e^{-\delta t} + (1-\gamma t)e^{-\delta t}\mathbb{E}_x[V(X_t) - V(x)] = o(t)$$

Intuition: The current value equals the expected discounted value over a small time interval t plus a possible dividend payment. In addition, γt comes from the first-order expansion of the Poisson probabilities (up to $o(t)$).

HJB equation

We find the change over time of the discounted part and the value function by differentiating $Y_t = e^{-\delta t} V(X_t)$.

$$dY_t = e^{-\delta t} dV(X_t) - \delta e^{-\delta t} V(X_t) \Rightarrow dY_t = e^{-\delta t}(dV(X_t) - \delta V(X_t)) \quad (24)$$

For $dV(X_t)$ we use Itô's formula. The dynamics are the same as in (19), but without the dividend term L_t. Set $a_s := \eta - (1 - u_s)\mu$.

Derivation of HJB equation

We consider two cases:

1. The company does not pay a dividend.
2. The company pays a dividend.

Consider the first case:

$$e^{-\delta t}V(X_t) - V(x) = \int_0^t e^{-\delta s}[a_s V'(X_s) + \frac{1}{2}\sigma^2 u_s^2 V''(X_s) - \delta V(X_s)]ds + \int_0^t e^{-\delta s}\sigma u_s V'(X_s)\, dW_s. \tag{25}$$

Derivation of HJB equation

Take expectations in (25).

$$\mathbb{E}_x\left[e^{-\delta t}V(X_t) - V(x)\right] = \\ = \mathbb{E}_x\left[\int_0^t e^{-\delta s}\left[a_s V'(X_s) + \frac{1}{2}\sigma^2 u_s^2 V''(X_s) - \delta V(X_s)\right] ds\right] \tag{26}$$

Let's do the same for the second case.

Derivation of HJB equation

$$e^{-\delta t}V(X_{t-}-\nu)-V(x)=\int_0^t e^{-\delta s}\Big[a_s V'(X_s)+\tfrac{1}{2}\sigma^2 u_s^2 V''(X_s)-\delta V(X_s)\Big]ds$$
$$+\int_0^t e^{-\delta s}\sigma u_s V'(X_s)\,dW_s$$
$$+e^{-\delta t}\Big(V(X_{t-}-\nu)-V(X_{t-})\Big). \tag{27}$$

Taking expectations yields

$$\mathbb{E}_x\Big[e^{-\delta t}V(X_{t-}-\nu)-V(x)\Big]=$$
$$=\mathbb{E}_x\left[\int_0^t e^{-\delta s}\left[a_s V'(X_s)+\frac{1}{2}\sigma^2 u_s^2 V''(X_s)-\delta V(X_s)\right]ds\right]+$$
$$+\mathbb{E}_x\left[e^{-\delta t}\left[V(X_{t-}-\nu)-V(X_{t-})\right]\right]. \tag{28}$$

Derivation of HJB equation

Substituting (26) and (28) into (23),

$$\nu\gamma t e^{-\delta t} + \mathbb{E}_x\left[\int_0^t e^{-\delta s}\left[a_s V'(X_s) + \frac{1}{2}\sigma^2 u_s^2 V''(X_s) - \delta V(X_s)\right] ds\right]\gamma t + \\ + \mathbb{E}_x\left[e^{-\delta t}\left[V(X_{t-} - \nu) - V(X_{t-})\right]\right]\gamma t + \\ + (1-\gamma t)\mathbb{E}_x\left[\int_0^t e^{-\delta s}\left[a_s V'(X_s) + \frac{1}{2}\sigma^2 u_s^2 V''(X_s) - \delta V(X_s)\right] ds\right] = 0 \tag{29}$$

Derivation of HJB equation

Dividing (29) by t and letting $t \to 0$ yields the stationary HJB equation:

$$\max_{0 \le \nu \le x,\ 0 \le u \le 1} \underbrace{\mathcal{L}^u V(x) + \gamma \left[V(x-\nu) - V(x) + \nu \right]}_{\Xi_x(u,\nu)} = 0, \tag{30}$$

where $V(0) = 0$ and

$$\mathcal{L}^u V(x) = \left[[\eta - [1-u]\mu] V'(x) + \frac{1}{2}\sigma^2 u^2 V''(x) - \delta V(x) \right].$$

Defining the barrier strategy

For each $\mathbf{b} = (b_1, b_2)$, define a periodic-classical barrier strategy $\pi^{\mathbf{b}} = (u^{\mathbf{b}}, L^{\mathbf{b}})$, where $u^*(\cdot)$ is the feedback control on $(0, b_1)$ (specified below):

$$u_t^{\mathbf{b}} := \begin{cases} u^*(X_t^{\mathbf{b}}), & \text{if } X_t^{\mathbf{b}} \in (0, b_1), \\ 1, & \text{if } X_t^{\mathbf{b}} \in [b_1, \infty), \end{cases} \tag{31}$$

$$L_t^{\mathbf{b}} = \int_0^t \nu_s^{\mathbf{b}} \, dN_s^{\gamma}, \qquad \nu_s^{\mathbf{b}} = \max\{X_s^{\mathbf{b}} - b_2, 0\}.$$

Defining the barriers

The optimal barriers are obtained by solving (30)

$$b_1^\gamma = \inf\left\{x > 0 : -\frac{\mu}{\sigma^2} f(x; v) \geq 1\right\}, \quad b_2^\gamma = \inf\left\{x > 0 : v'(x) < 1\right\}, \tag{32}$$

where $f(x; v) := \frac{v'(x)}{v''(x)}$ and $\inf \varnothing = \infty$.

Optimal strategies

Based on the optimal barriers, the optimal strategy is conjectured to take the form

$$u^*(x) = \begin{cases} -\dfrac{\mu}{\sigma^2} f(x; v), & \text{if } x \in (0, b_1^\gamma), \\ 1, & \text{if } x \in [b_1^\gamma, \infty), \end{cases}$$

$$\nu^*(x) = \begin{cases} 0, & \text{if } x \in (0, b_2^\gamma), \\ x - b_2^\gamma, & \text{if } x \in [b_2^\gamma, \infty), \end{cases}$$

Illustration of the strategies

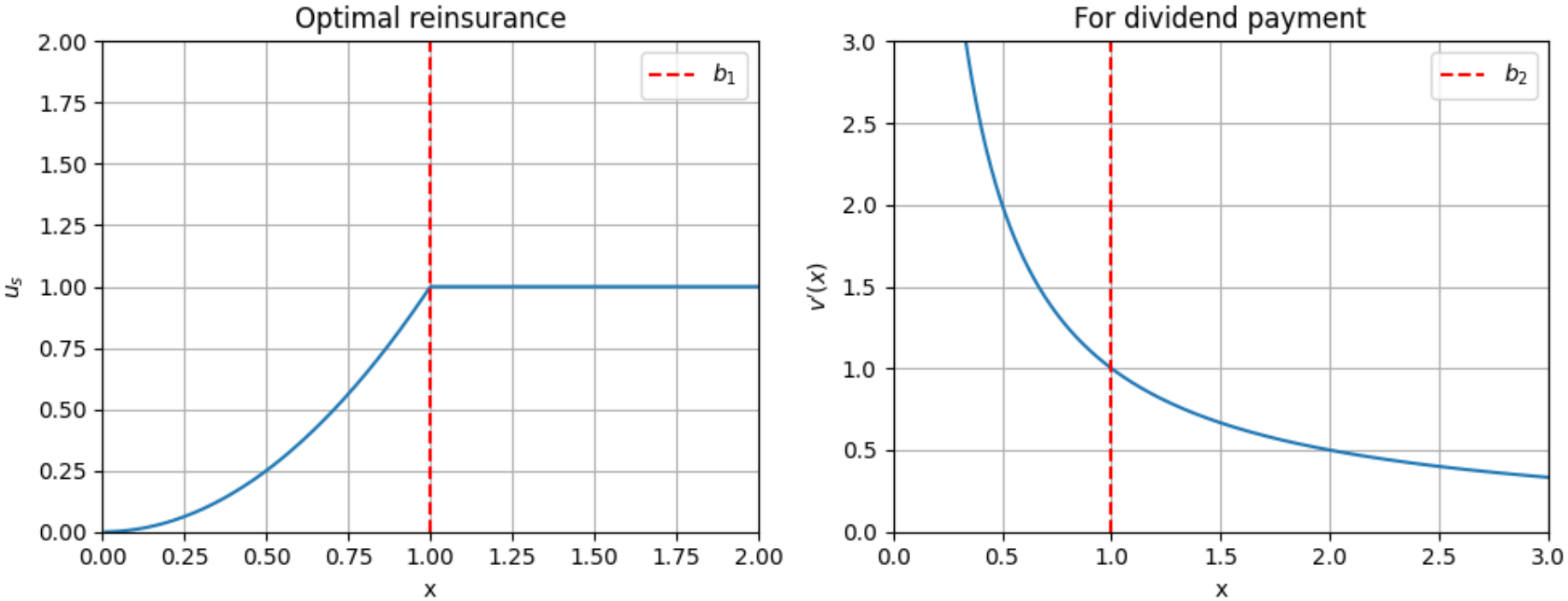

Figure: Illustration of the barriers

System of the differential equations for $v(x)$

$$-\frac{\mu^2[v'(x)]^2}{2\sigma^2 v''(x)} + [\eta - \mu]v'(x) - \delta v(x) +$$
$$+\gamma\{x - b_2 + v(b_2) - v(x)\}\mathbb{1}_{\{x-b_2>0\}} = 0 \quad \text{for } x \in (0, b_1), \quad (33)$$

$$\frac{1}{2}\sigma^2 v''(x) + \eta v'(x) - \delta v(x) +$$
$$+\gamma\{x - b_2 + v(b_2) - v(x)\}\mathbb{1}_{\{x-b_2>0\}} = 0 \quad \text{for } x \in (b_1, \infty), \quad (34)$$

$$\text{s.t. } v(0) = 0.$$

Solution for the very expensive case

$$v_{\gamma,b}(x) = \begin{cases} c_{1,1}(b_2)\, h_1(x), & x \in (0, b_2), \\ c_{1,2}(b_2) e^{\lambda_\gamma (x - b_2)} \dfrac{\gamma}{\gamma+\delta}[x - b_2 + v_{\gamma,b}(b_2)] \dfrac{\eta}{\gamma+\delta}, & x \in [b_2, \infty), \end{cases}$$

where λ_γ is the negative root of

$$\frac{\sigma^2}{2} r^2 + \eta r - (\delta + \gamma) = 0,$$

$$h_1(x) = e^{\theta_+ x} - e^{\theta_- x} \quad \text{for } x \in (0, \infty),$$

with θ_+, θ_- the positive and negative root, respectively, of $\frac{\sigma^2}{2} r^2 + \eta r - \delta = 0$.

Solution to the Cauchy problem

$$c_{1,1}(b_2) = \frac{\frac{\gamma}{\gamma+\delta}\left[1 - \frac{\eta\lambda_\gamma}{\gamma+\delta}\right]}{h_1'(b_2) - \frac{\delta\lambda_\gamma}{\gamma+\delta}h_1(b_2)}$$

$$c_{1,2}(b_2) = \frac{1}{\gamma+\delta}\left[\delta c_{1,1}(b_2)h_1(b_2) - \frac{\gamma\eta}{\gamma+\delta}\right]$$

Assumptions/Limitations

- In some cases solving the non-linear differential equation analytically is impossible.
- For some processes it is quite hard to solve the optimization problem (i.e. compound Poisson process).
- Formulation of HJB equation requires rigorous math.
- Non-linear differential equations can be solved with PINNs (Physics-informed neural networks)

References

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